ActInf Textbook Group ~ Cohort 4 ~ Meeting 10 (Chapter 4 part 2)

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right
greetings everyone Thanks for
joining we're on eight eight in our
second discussion on chapter four
so
one question was uh brought up that will
definitely come to but is there just
anything else that anyone else wants to
add about for
any just reflection or any question
that's kind of surfacing that we want to
add to the stack
all right let's go to the message
passing
okay
so in message passing is there a decay
of information as the distance between
the variable X and individual Markov
blanket constituents increases
is implementation of information Decay
and message passing an option for model
implementation
who wants to give a a first thought on
that
or any any other kind of curiosity
around it
um
so would it
I'm not sure if I'm understanding it
very well other than intuitively but
would it I mean would it be the case
that uh like information Decay wouldn't
necessarily
um I suppose makes sense here given the
the models they're already working

uh in in terms of like time steps and so
you're already
kind of um
you know you're you're recomputing
uh your your gradients and your
posteriors and so on in every given time
step
such that
um you know the idea of some kind of
uh holistic piece of information that's
being carried
uh from time step to time step doesn't
quite making sense that is each time
step involves an entire set of like
recalculations does that make any kind
of sense
yeah
um
so just to start with one uncertainty is
I'm I'm not exactly sure about message
passing in the continuous versus
discrete time setting
um again hopefully something that we we
can unpack and talk to Magnus kudal at
all since I know that he's an expert on
the continuous time
um setting
but
any other thoughts on this or I'll I'll
try to give uh
an answer Ollie
uh yeah just one quick note about
message passing is I mean before going
into the details of how exactly message
passing happens in active inference
framework uh it's probably worth noting
that uh it's inversely proportion
proportional to the robustness of any uh

self-organized
self-organization systems because
you see uh if we consider an agent and
uh it's um I mean peripheral
um systems or peripheral
um let's say limbs as a kind of central
like fully centralized system uh in
which uh I mean the message passing
happens uh almost uh in a fully
centralized way from uh the brain or I
don't know the central processing unit
onto the limbs then this system would
not be as robust as the system in which
uh this kind of message passing is is
distributed among
um among the elements of the system so
I think that's one of the reasons why we
see uh signal Decay or I mean the loss
of information
as we go up to the hierarchy and uh I
mean the more we go up to the hierarchy
the less precise the information becomes
uh but in order to have more and more
precise information we need to somehow
acknowledge a kind of autonomy onto the
peripheral
um nervous system as opposed to a
central nervous system so uh that's one
of the ways uh I mean in humans and
nervous system is organized in which
this kind of distribution of uh agency I
mean
autonomous
agency is distributed among both
peripheral system and the central system
so
[Music]
um

and coming back to active inference
framework
um I also believe in this framework it
beautifully captures this kind of
hierarchical model because obviously as
we move farther away from the peripheral
elements of any system
the elements which have direct access to
the direct access but closer access to
environment then the the accumulation of
probabilistic Errors becomes larger and
larger and that's why
um
in order to have a reliable information
we need to discretize these kinds of
information at some point without uh
necessarily having to take it into
account every single bit of information
that we receive so uh
on the lower levels of the uh I mean
when we're dealing with the lower level
systems we mostly
deal with continuous time models
which allow for more precise kind of
formulation of message passing but as we
go up to the
central nervous system and concede
discretized elements of organizations
such as cognition emotion and so on or
decision making this process becomes uh
much more discretized so that's
basically the idea behind the Hybrid
models of active inference in which we
can incorporate both continuous time and
discrete time elements into a single
system
that's awesome thank you ollie Alexi
I wonder if I can ask you Ali to clarify

so uh let's imagine that I'm a shark
right and if I go up the hierarchy of
beliefs then perhaps the most uh firm of
course green belief is that I Will
Survive
one step down from that is that I can
breathe in water right so
if as the message comes up in the
hierarchy I don't need as many details
like
um because these are not going to change
if they change they die
is that kind of what you're saying that
exactly no exactly yes because as you
said as we go up to the hierarchy where
we'll need a much much less detail and
we just need to take into account uh I
mean the general gist of the information
we get in order to survive or I don't
know achieve a goal or something but
obviously
um if if we want to I mean take into
account every single detail of every
Endeavor will need much a more precise
channels for information passing
and if I made one more quick question in
chapter five they make a point to say
that uh um ascending messages which are
prediction errors are usually higher
frequency than descending messages I
mean they declare it but I didn't
exactly see where they take it from and
so I was trying to uh they Illustrated
with a cortical column and I was
thinking that you know in indeed this is
very interesting research that we have
top-down messages that are you know
better written or something like that

and and uh they govern the gamma Which
is higher frequency could you talk a
little bit about why prediction errors
and ascending messages are higher
frequency non-linear functions
okay uh I mean on this question I can
only take a guess because uh I don't
honestly know the exact answer but uh I
believe the reason that the ascending
information is uh I mean
um they're on a higher frequency level
than
um I mean the descending information is
because
again coming back to this hybrid model
of continuous time and discrete time
ascending information kind of deals with
um let's say a continuous discretization
of information
and that's why as we go up uh in the
hierarchy and the hierarchy
will need to have a much more uh I mean
a much higher frequency for the
information gain because we need to uh I
mean we need to construct those discrete
information
uh as quickly as possible and based on
those discrete elements of our
information we can then
infer any necessary prediction any
necessary information about the error
that we get or any other kinds of
manipulation or optimization uh we need
to take into account and I believe this
kind of
bi-directionality uh Maps quite uh
well on adaptive resonance Theory
grossberg's adaptive resonance Theory

because in that theory as well this uh
kind of bi-directionality between the
ascending information and descending
information comes in a kind of resonance
and
um they adjust their frequency as to
become the resonance uh fully cyclic
passage for information both going up
and going down so that's why I believe
in this kind of framework also it's
necessary to take into account different
rates of frequency for both ascending
information and descending information
in order to allow for this kind of
resonance to happen between
um
those two lanes
this this is awesome it's a lot of a lot
of great information
um
I'll just try to add a few pieces
because I think there's different
aspects here
um so first I'll let you pointed to the
the topology of message passing
messages are being passed on the
topology of the variables so how the how
the variables are connected
is how the messages are being passed
sounds kind of circular but that's the
whole reason why this box
discusses basically the markup blanket
which is just describing any given node
is going to be surrounded by a blanket
for that no there's an absolute tag
on a given node so it can't be said just
singularly this node is an internal
state or a blanket state only with

respect to the whole partition but
that's why two message passing
strategies
um are being described directly
alongside the Marco blanket so a lot of
times we see that particular partition
inactive inference or just more broadly
we see a Bayesian graph and how the
variables are connected
and this is going to be an
implementational or operational
procedure
that's gonna
compute
what it's going to turn the crank one
gear for a given
topology of connectivity
um so this point about the robustness is
similar to to the robustness of a
distributed system I mean if you have a
fabric with only local connections then
it may take some time to propagate or
percolate with the information
but on the other hand there's not a
single note of failure and all these
other
um features
um
message passing graphs can include
discrete and continuous time and
continuous State spaces like all of
these different variants that we've been
discussing play nicely together chapter
eight comes back with a hybrid model and
livestream 46 on the folks psychology
where in the periphery it's more like
continuous sensation and action and then
as you um get towards like the cognitive

decision-making States you're dealing
with discretization
um okay
uh another point is about like the
computational tractability
so there are um works like this one from
2019 with uh par at all
where
let's learn more about the technical
details but basically there's a few
different flavors of this message
passing implementation
that are also kind of hinted at in the
um in the Box
but suffice to say that like for a given
topology of a Bayesian graph which might
be describing a cybernetic system active
agent or it could just be any other
Bayesian graph this isn't about
perception and action specifically
but on any base graph
you can basically Define an
implementational procedural pseudo code
for how to just
bring it that much closer towards
harmonization
where if it's kind of the ball is
already at the bottom of the bowl it's
not going to go anywhere but then if
it's not at the bottom of the bowl from
like a variational inference perspective
it will go there and there's a procedure
for doing that that's that's tractable
um
why is the frequency higher on ascending
messages than than descending so first
these kinds of phenomena
constitute empirical evidence for

hierarchical predictive processing
architectures but but the framework
doesn't depend on these phenomena being
the case for our real nervous systems
but but they're always nice to find
and there's a few ways to go here
um one connections with the Nyquist
frequency
so this is um if you're going to be
sampling from Vinyl from analog
then that's a smooth curve that's the
continuous time that's like the motor
perception
that's the peripheral nervous system
that's the record
that's like a clock with a smooth moving
minute hand
and then
you're going to have it it translate to
digitization like you're digitizing the
record
um so if if um we're get if there are
top down predictions like about the hour
in the clock uh metaphor
we could have ascending predictions
about minutes and descending predictions
about hours happening every uh clock
cycle
there's no reason why you wouldn't do
that but at the same time you can you
can bake in
this structural understanding that hours
are top down and slower
which is consistent with more broadly in
nested modeling nested outer layers
um deeper however you want to think
about it like they are slower and bigger
so you only need to send down that hour

uh expectation
more infrequently
and then just lastly like signal decay
um if you have a
a Markov chain so just think really
um simply about some
uh just just
without bringing action too much into it
think about just the perceptual part
here so just like the music listening
example
um that I think is in in um so here here
we are this is a partially observable
Markov process
happening through time emitting an
observation so it could be the
temperature changing in the room Through
Time
but we're not intervening so no action
yet
um imagine if B
was if B worth the identity Matrix like
all ones on the diagonal you would just
be carrying s forward
with perfect fidelity
but if B were like 0.7 on the diagonal
and then point one on the off diagonal
which is like what the case that it
wasn't a music example
then you can if you set up a hundred of
those runs next to each other it'd be
like the signal was kind of like
blurring by 10
um each time
so
the signal it it depends on like if
there's a degradation of a signal as
it's being passed here it's being passed

in time
if it's being passed noiselessly then
the signal propagates
um
more distantly in space and or time
whereas if um the the insulation around
a node is such that it it
um blurs half the information upon its
immediate exit then that's like then
that information
like that perturbation the base graph
um
it just dampens
more rapidly
Ali
uh thanks staff connection to Nyquist
frequency was really insightful and
helpful but uh to add at your point here
I also believe that if uh if there
wasn't any gradation of frequency uh
between those two Paths of information
passing I believe uh something like an
aliasing effect would happen uh if if
the brain or any uh kind of agent uh
wanted to reconstruct the information
from the input system and then produce
the necessary output system because
um I mean it's necessary to
somehow uh distinguish between the kinds
of uh I mean the frequency of incoming
information as opposed to the uh the
frequency of the constructed information
uh in order to evade these kinds of
aliasing type
effects which obviously would be
undesirable
yeah cool and also ties to the um
ultimately the audio visual

roots of predictive processing it was
not generated to describe or explain
biological phenomena it it arose in the
70s
20 years before round Ballard 1999
in the context of Maximum information
compression for Audio Visual
and then the the other
um sort of connection back to the more
like bodily
maybe a book that that some of you are
are more familiar than than I but the
body has a head
it's kind of like a exploratory
narrative
um biology physiology book
but but of course the title is the
provocation about the body's Supremacy
in I guess what we would say action
inference Alexi
yeah I wanted to reflect on what you and
Ali said um about the Nyquist frequency
and also that going back to please
forgive me ascending messages uh uh
prediction errors higher frequency if we
take this uh evolutionary kind of
thought that
uh you know the higher in the hierarchy
I go the more difficult it is to update
the prediction they're more well
entrenched right so let's imagine that
I'm a rat but I have an inborn phobia of
cats in a one strand of cat hair you
know because it's freezing but if I'm in
the environment where
there's mortal danger like a trap and
the only Escape Route smells like cats
then I have to update that very firm

prediction to get out
so it takes high energy which in like
Planck's equation means high frequency
to update the high level predictions I
know it's very speculative but what do
you think about that
yeah I think there's a few ways that
that you could model that
um like we let's just say we we have a
um we have an aversion to cat smell
um like it's a negative value in C or
it's a lower value in the C Vector just
to be clear like we're talking about
some sensory Vector
um and there's in our generative model
we have some value that just is called
cat smell and we prefer
it to be lower
um well if there are two paths on the
maze
then
to take the path with the cat smell it
it has to have pragmatic or epistemic
value
so if we have a ten to one
um aversion to the cat smell
but then the expected free energy of the
cat path
is greater than ten to one
then
you could think about that as like the
circumstances
um reflected by the epistemic or the
pragmatic value of a course of action
but the circumstances would lead to
um selection against that
um there's probably other ways that it
could be explored um one would yeah but

um yeah higher or so and and uh there
may or may not be these kind of
it may or may not be in
option like I'm thinking about trying to
try and just like trying to see two
colors as indifferent when you can
clearly see them as different or
something like that like you're gonna
butt up against
some fundamental
like
the wavelength of light or like maybe
there's a sound that maybe you know some
sounds are maybe culturally aversive or
not or Pleasant
but maybe there's also just a volume or
some type of sound that that it's like
yeah probably you can't can't
but then again if that was the only path
to escape a burning building
then that kind of stuff would happen
so then I think it you gotta zoom out
beyond the single
dimensional generative model
and then consider more holistic
generative model like we have ten to one
preference against cat but we have a
hundred to one preference for blood
glucose
so then right there you've constructed a
generative model that enables you
to make risky decisions
thank you that's very very useful yeah
yeah it's
uh and in the end it's it's not about
the rat that's the territory it's about
the generative model
so different modelers May construct

different generative models
that display certain kinds of
um Behavior or not so we have the
portfolio of models
and then we ask well what experiment is
going to help us update our attention to
these different models
and someone has one where there's only
you know cat smell yes or no somebody
else has no a little bit and a lot
somebody else treats it like a
continuous variable
none of those are the true map but but
we have all the maps on the table
and then that allows us to design the
experiments
but for the circumstances that would
differentiate the unique predictions and
explanations from those Maps really
it also reminds me of Terence Deacon's
uh nest of conceptions of information so
instead of representing these kinds of
relation hierarchical relations as a
higher order or lower level uh lower
order processing and information
he represents them as three nested kinds
of information in which the Shannon
information is would be the most basic
one and then
referential information which is um
uh he also refers to as aboutness
information
uh incorporates a kind of medium
susceptibility to any error and it is an
emergent information rising from those
Channel information and ultimately we
come to significant information or in
other words information about usefulness

which would which is what actually
dictates how the survival of the species
uh ultimately is determined uh I mean
through the evolution and so on and it
it incorporates a kind of
the
connection or association between uh the
reference of that usefulness information
and the teledynamic requirements of this
species is what dictates what would
happen in the significant information or
in terms of fep uh the minimization of
free energy
nice that's that's awesome yeah
information is like one of the most
overloaded terms or just loaded terms
but
um Shannon 1948
studying signal Transmission in the
telegraph Networks
so it has to do with that pure Morse
code level of being able to distinguish
like a signal from noise
and and then projecting all that down to
the Shannon entropy
and then
to the extent that a signal can be
detected from noise
which is about the Observer not the um
the system in itself already then
there's this like semantic or symbolic
information
but symbolic or semantic information if
we're gonna
um continue with this evolutionary line
like okay so here are the photons we can
pick out the signal in the photons a
signal and photons

it looks like it's yellow and red
is that a good caterpillar to eat
so
one has to go beyond the kind of Shannon
information if you just had the entropy
Channel entropy of different natural
settings as a bird
that's still
um two structured jumps away
from like
should I go for this caterpillar or not
but a lot of times this is taught as
information Theory kind of the beginning
and the end of information Theory when
this is the syntactic information and
then the semantic but the the the
abstract semantic is still not the
actionable information
um
let's say someone has added the key
things how you define the markup blanket
uh
I mean yeah but we're not playing with
the definition of what a Markov blanket
is maybe this could be rephrased as the
key thing is around which node you
choose to highlight as your system of
Interest call it internal States and
then the Markov blanket is trivially
defined around it
I.E there's a partition between whatever
or there's no partition well again
if they if you're using a plural
there is a partition between them that's
what makes you think that they're two
different things
and everything under the or there's no
partition and everything out of the

blankets conditionally dependent well
the fact that you have modeled a blanket
or a partition at a given scale doesn't
preclude there from being as they say
message passing within the same blanket
because blankets are not something
fundamental it's like saying
um could messages be passed inside of
California
yeah
but but then also you could go to the
county level
and there could be messages between the
counties or between the cities or you
know between the agricultural objects
within the the region
I've not seen this with love if somebody
putting references I'm not sure exactly
which exact reference but more broadly
nested
um Markov blankets
could look to the kind of Earth Systems
modeling or
our uh variational Eco Evo Devo paper
nested markup blankets they're having
message passing within and across
and the kind of like lateral same type
of thing same scale message passing
and then now that you're you know you
you passed all the messages amongst the
states and now we're in California and
then you pass all the messages amongst
the counties and then but I don't have a
specific reference where that is is
written out
but yeah what an interesting question
and and message passing is so cool
because like it really connects

like this
but what if this was just an abstraction
and there wasn't any procedure known for
how to compute it
then you'd have kind of like the
blueprint of a building
but
it'd be awesome if you could build it
but in fact this is the blueprint
and the blueprint can basically be
rendered
or operationalized into a procedure
so that is is a super super strong
connection and also one that's not
unique to active inference
um although
i i as far as I know
um
the the 2017
this paper I I believe is far more
General so somebody can you know
add more here but basically
um
here's the continuous and the discrete
representations
and uh
in this paper and some or some other one
from this era like they showed that
every base graph
has a Forney Factor representation
and then previously it had been shown
that every 40 graph had a message
passing implementation
so in this paper or some other one
um like it kind of connected all the
pieces up and said yeah now there is a
message passing procedure on arbitrary
Bayesian graphs

again that doesn't have to be about even
a perception Action System
I'm I think it just made the general
case
um
some message passing is definitely
important and also it's another example
of like uh perhaps even a difference in
practice between uh like deep learning
reinforcement learning machine learning
Ai and active inference to the extent
that the distinctions can be drawn
um in uh just to highlight because
there's of course multiple differences
that we've talked about but a lot of the
work in Ai and ml goes into like
figuring out the implementational detail
like kind of getting
um getting in there with the specifics
of the model
whereas
once something is stated as a base graph
then
it's kind of like
standard methods
like you don't need to intervene in how
your source code is compiled
and so similarly that the message
passing step kind of compiles it into a
procedure
but you don't need to step in to to
fine-tune how the blueprint goes to the
construction
which I believe is a great advantage
anyone want to want to go to a specific
question or we'll just try to look
there's so many questions on on chapter
four so anyone can just like bring up

one or or some topic that they thought
was cool
in the chapter
um actually I did want to briefly just
kind of harken back very quickly to
something that Ali brought up earlier
um which is uh adaptive resonance Theory
and uh maybe how it relates to
the asymmetrical frequencies found in
message passing
um and I know in chapter five where we
start discussing kind of uh or we start
seeing more discussion of uh like
neurobiological correlates and things
like that
um you know there are these references
to um
um non-linear functions being involved
in the competition for dictionaries such
that what is it a ascending messages
tend to be much higher in frequency or
faster than
um descending messages I mean are all of
these things kind of
kind of related uh you know as far as
message passing goes
I'm rather into things like fmri and EEG
analysis and so I I kind of have a
personal interest in uh the asymmetrical
frequencies question
um if all of that tracks
yeah
well I think that the uh Sensor Fusion
neuroimaging world
is is where a lot of these
statistical methods can be traced
um
by virtue of the SPM Origins

and here this
um live stream
was um excellently prepared for by Ali
uh Professor grossberg was one of the uh
relatively small fraction of guests who
prefer
full cited referenced questions before
joining but also he prepared excellent
answers
and there was a great discussions
um
about the compatibility or or
reconcilability of of Arts fep active
everything
Lexi
yeah I'm just grateful for your question
again going back to it and I have the
same interests but
um you know come to come join us on
August 23rd if you can on uh the cash
Theory paper I mean I think I'll try to
talk about how the higher the frequency
of the EEG the higher the common good of
Sinai entropy the higher the level of
keraticity and there's much evidence
kind of converging on that stuff today
on Twitter of all places uh Earl Miller
from MIT was talking about uh better
frequencies uh governing uh the exchange
between medial prefrontal cortex and
right hippocampus for spatial uh work in
memory so it that does seem to kind of
not only be theoretical but also
confirmed empirically that we have kind
of uh control signals are slower think
about the brain you know uh subcritical
neurons are firing one at one tenth of a
frequency of cortical neurons and

subcortical neurons and brain stem are
arousing the cortex they are kind of the
blood that and the energy the arousal
generalized arousal comes you know that
way uh and another thing I wanted to
mention is about the prioritization uh
of sort of
um Mark Songs talks in his book about
how uh working memory is a very small
space like seven chunks of information
plus minus two and when we feel
something it has to get into work in
memory otherwise you know we have to be
conscious of an intense feeling and
ordinarily for example we don't like
urinate or defecate in public this is
embarrassing so we hold it we inhibit
the wish
but when people go through like public
execution or hanging that wish is
completely released because there is a
higher priority there's an existential
catastrophic threat and that this sort
of uh fear of social embarrassment is
completely deprioritized so it does
suggest that we need a very high energy
signal to kind of overcome very well
entrenched predictions so it's a vast
topic and I'm highly interested in it so
uh if you're interested maybe we can
talk and collaborate
cool
cool thank you and again to connect that
to the Terence deacons like a high
energy signal
are we talking about an informative
signal because an informative signal
could just be one zero one zero one zero

one zero cleanly or something like that
are are we talking about like a powerful
like meaningful you know like the just
some some cultural Landmark that has a
specific aboutness but isn't
situationally relevant
now the high energy is is happening at
this higher cognitive level
might even be be
um uh
uh subtle
signal informationally or it might even
be a subtle signal symbolically or
semantically
but that's the whole thing like early
warning systems and like you know just
the movement of of a shadow or the Rope
coiled up on the ground might convey
extreme actionable information
[Music]
whereas there might be something in a
different quadrant that's like very
clear but not
um survival relevance
exactly I mean I think that if you go
into like ecological evolutionary goal
of mammals to survive and reproduce then
okay so survive means eat and don't get
killed and so uh sexual functioning and
uh gastrointestinal functioning is
disabled when we are in a state of acute
fear acute fear takes precedent because
if I'm dead it doesn't matter you know I
will not be able to reproduce so my
first job is to there's a prioritization
happening and that's going to be an
intense single that just you know uh
occupies the entire mind and just casts

everything else out right
um yeah that I mean a lot of these um
whether we approach it from like a
circulatory system or like a more
abstract information like a lot of these
um patterns are are explained or
predicted or or so-called paradoxes are
are um kind of shown to be otherwise
one's like yeah you can't have Max blood
flow to every region at the same time
like if you if you could do that then
you would just have some higher level of
allocation to do so yeah when you're
running your gastrointestinal tract
slows down like
there's just a shunting of resources
and there's a shunting of attention as
well so it it doesn't make sense
to
um have a system where equal maximal
attention is being
deployed
it it it's just not viable but
we can describe it
so that's like why the the active motifs
and and all these like like
Concepts these are our grammar or or
ontology
that gives us the expressivity that we
need to do the modeling
but
um so we can describe systems that yeah
maybe there is a system that pays equal
attention to all of its 10 ideas
but or not
or you know it it doesn't have any
ability an organism that doesn't have
the ability to shunt

um or redirects
uh blood flow or attention
or does it doesn't have the ability to
do sensory attenuation or something I
mean you can build it
okay a lot of these questions are about
um
specific
um like variables
we may have come to this one last week
why is the marginal likelihood called
marginal
does anyone
um
know this or have a thought
yeah I'll leave go for it
it's a Remnant to the statistical Theory
because
um you see when we sum the probabilities
of a of a given event uh in in a tabular
form uh the the ultimate sum would be
written in the margin of the table so
that's where this marginal probability
term comes from
perfect yeah what a great what a great
term it's in the margin of the
spreadsheet
when they summed across
um
kind of collapsing on one thing
conditioning on one thing happening
you can collapse across the other
dimension on the spreadsheet and then
it's in the margin
um yeah so
so many
questions
um another another kind of uh

uh clear one
uh
belief updating about policies
two update beliefs about policies
we find the posterior that minimizes the
free energy does a posterior at time T
become a prior at time t plus one
yes it does so like d
we we call that variable in the
generative model
the prior
because it gets the whole chain rolling
but you could also think of this as just
like kind of being like a starting
position on this unfolding sequence so
here's a temperature in the room and
we're just going to say well our data or
our modeling started at midnight
and it was
um 30 degrees
we're just going to inherit 30 degrees
if it was a precise prior or you could
say we're going to start the chained by
pulling from a prior distribution
so we're going to start the chain with
30 plus or minus 10. and then we're just
going to let it play on and then in that
situation you know with respect to the
presence moment S Sub t
um this is a posterior with respect to
what just happened
but it's a prior for what is about to
happen
so just like Marco blanket is not kind
of absolute
it's about which variable you're talking
about also prior and posterior
are relative to the incoming observation

so one
after you've updated the posterior it's
not like you just stop then that updated
value for the hidden state
becomes the the the prior
for the next incoming piece of of
sensory data
a lot of equations that I hope we can
unpack and develop in in math group and
and otherwise
um and and like annotate richly and make
sure that for all the equations that
we have those
um descriptions
and yeah chapter fours
a big one
I mean there's a lot to it and
brings up a lot of
big topics
that was our second discussion on it
we are close to
the end
we'll go into chapter five next
also I remember thank you Lexi
Bronwyn I think you
some discussions that we had one of the
specific
um message passing neurobiology in the
setting of chapter five so how should we
prepare ourselves
yeah that was a while ago
yeah it's a tricky subject I've been
listening and thinking
um through today
I just get I've just gone back to um
equation 2.6
um because I got a bit confused
um

but it was just about
um you know message passing in the
relationship to uh the reduction of free
energy and expected free energy
but I'm still thinking about it so I
might discuss it next week if I come up
with something
but I think it's um
it's very complex and then you've got
the hierarchical thing happening but but
one question I had was that over a
synapse there's sort of two functions
one is a facilitation one's an
inhibitory process and I think we don't
consider the inhibitory process
our ability to inhibit
very much it's sort about all about
action
that um
yeah I have to think about all that but
that's um I mean in this yeah in that in
that equation you've got all of your
um
all of what you were talking about
taking risk
um you know that that thing about the
cat the cat smell
example
but within that selection of policy for
the reducing expected free energy
um you want the one with the least
expected free energy but you're also
balancing up
other factors in terms of
the the three nests I think the three
nested thing was really interesting
because it gave you that that that
hierarchy of Habitual responses as

opposed to
um considered ones and you know higher
referential or significant um
evolutionary considerations
but underneath it is this is this
equation
I think the relationships between these
things in that in that moment when
someone makes a decision
so
yeah it's quite a complex thing that the
brain does really
that's my thoughts
very basic but that's where I sit so
anyway whatever I've been doing other
things lately so I'm a little bit
um
I lost and I haven't had a good look
re-look at chapter four that I will
between now and next time
awesome yeah a lot of ways to go there
this
um functional this this um can apply to
a nested system it's just all kind of
abstracted there
and again this is kind of like that
blueprints view message passing helps us
get there
with the how so it's kind of like the
low road which is where this equation
comes from chapter two yeah this it's
it's like it's explained the how and
then the message passing is like the how
on the how
and then also like what you said about
um you know one one uh world view is
like the central uh operator Central
Governor is kind of being like spurred

into action it's all about like
excitatory signaling from the outside
um or um it having to stimulate a
passive exterior through like um you
know carrot and stick or just stimulus
in general
um and then that reminds me of Tom froze
um with his guest stream on the eruption
which is like yeah what if it's like
there's there's just this like seething
activity and then what's happening is
like inhibition and and breaking
and then like in those spaces that are
opened up through inhibitions and breaks
things unfold
so it's not like the decision making was
was like well you know shining the laser
on where to go it's like no there's
there's more than enough energy to go
around mental energy and it's just about
um knowing when Once inhibit went to
break yeah absolutely
I think I have a problem the problem is
a season that's knowing when to inhibit
and that's that's from the work that I
do really is a strong point
um of of
inhibition as a very important part of
decision making
an action in the world just how we
behave really I mean you know if you
look at the basis of a synapse if you're
in facilitating or inhibiting and that's
the balance between those two in terms
of message passing
but nobody talks about inhibition very
often
nice yeah yeah great

I mean in in the um number go up world
is good more reward is better stimulate
reward reward is stimulating you know
sugar is good all that versus in the
predictive processing at least we open
the discussion to to both hands
uh of up and down and
um stimulate and inhibit or maybe
neither stimulate nor inhibit just
different
yeah so it's it we have all the axes we
need and and that's why chapter six
exists because so much of the challenge
is like
what building do you want to build
so if we said okay well from this
we can optimize it super tractably it's
not going to be an issue
but what is it what is that model going
to be
and how are you going to iterate on that
in a real setting
which brings the whole
um understanding into
action and earlier
um
yeah this was just the last paragraph of
of uh the book
active inference is not something that
can be learned purely in theory
no it's the action people keep
forgetting about the action the two are
entwined you can't separate any of it
uh so yeah awesome
yeah well
fun fun discussion thanks Andrew for for
highlighting the message passing
um

angle and it's going to be a perfect
lead-in to
um chapter five
where we will revisit message passing
again from this like nervous system
ankle yeah
right okay thank you until next time
thank you thanks everyone so much fine
bye